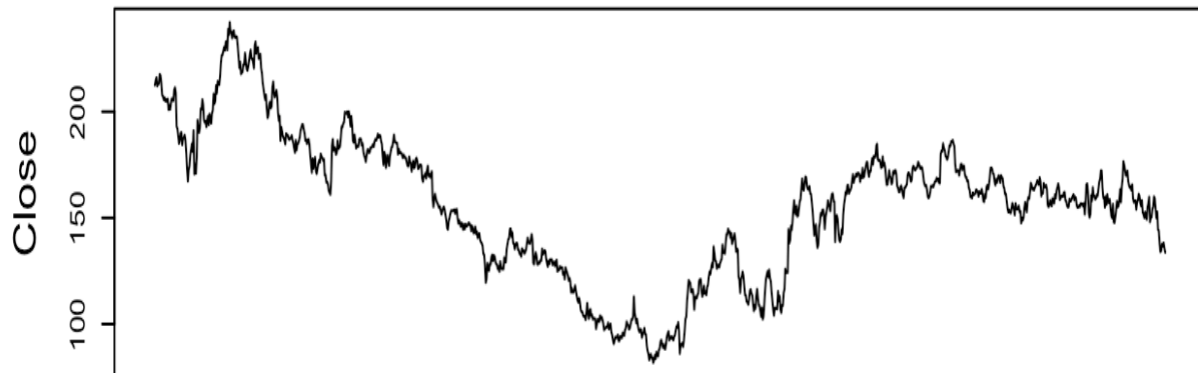
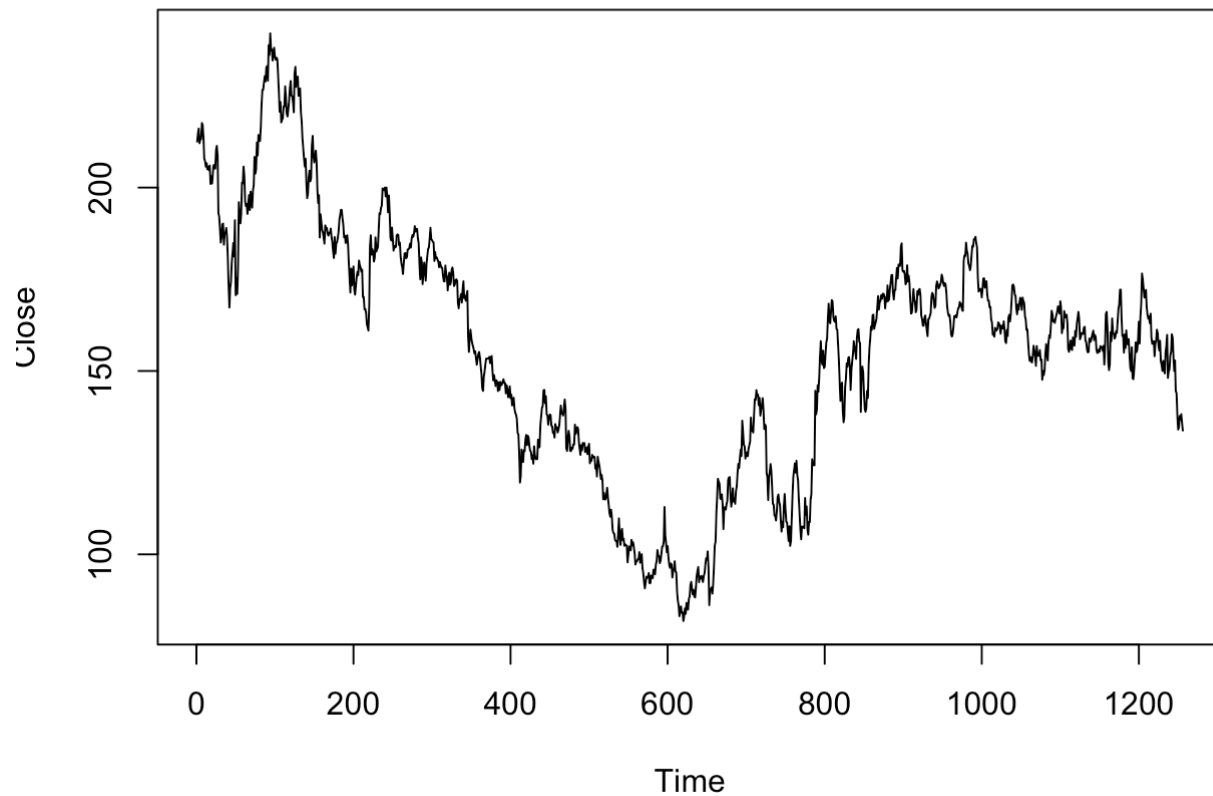


### Summary - Amazon Share Price Data for last 5 years (finance.yahoo.com)

I started using the same Amazon data that I used for Matlab for doing analysis in RStudio. After performing some analysis on Amazon closing stock prices within RStudio, I was able to find that our model produced an ARIMA(0,0,0) model which represents white noise (page 5). White noise practically means that the data is random with no trend being displayed. The model signifies that there is no specific relationship between the variables and the data which means that the time series is random. Due to the white noise, I stopped working on the Amazon data series and moved to the Bitcoin data series and that is what the project is based on.

Time Series Plot:





Code:

```
plot.ts(Close)
```

Code:

```
Amazon_Stock_Price_Data_Sigma_Scroll <-  
read.csv('Amazon_Stock_Price_Data_Sigma_Scroll', header = T, sep = ',')  
Amazon_Stock_Price_Data_Sigma_Scroll <-  
ts(Amazon_Stock_Price_Data_Sigma_Scroll, frequency = 12, start=c(2001, 1))  
Amazon_Stock_Price_Data_1_1_  
plot.ts(Amazon_Stock_Price_Data_Sigma_Scroll)
```

Link:

[▶ Plotting a Time Series graph on R Studio](#)

Step 1:

```
attach(FILE NAME)
```

Step 2:

```
library(tseries)
```

Step 3:

```
library(forecast)
```

Step 4:

```
plot.ts(Close)
```

Stationary (Dickey-Fuller Test): Test done in stats to check if a data set is stationary

Code: `> adf.test(Close)` - Check for stationary

Augmented Dickey-Fuller Test data: Close Dickey-Fuller = -1.8452, Lag order = 10,  
p-value = 0.6439 alternative hypothesis: stationary

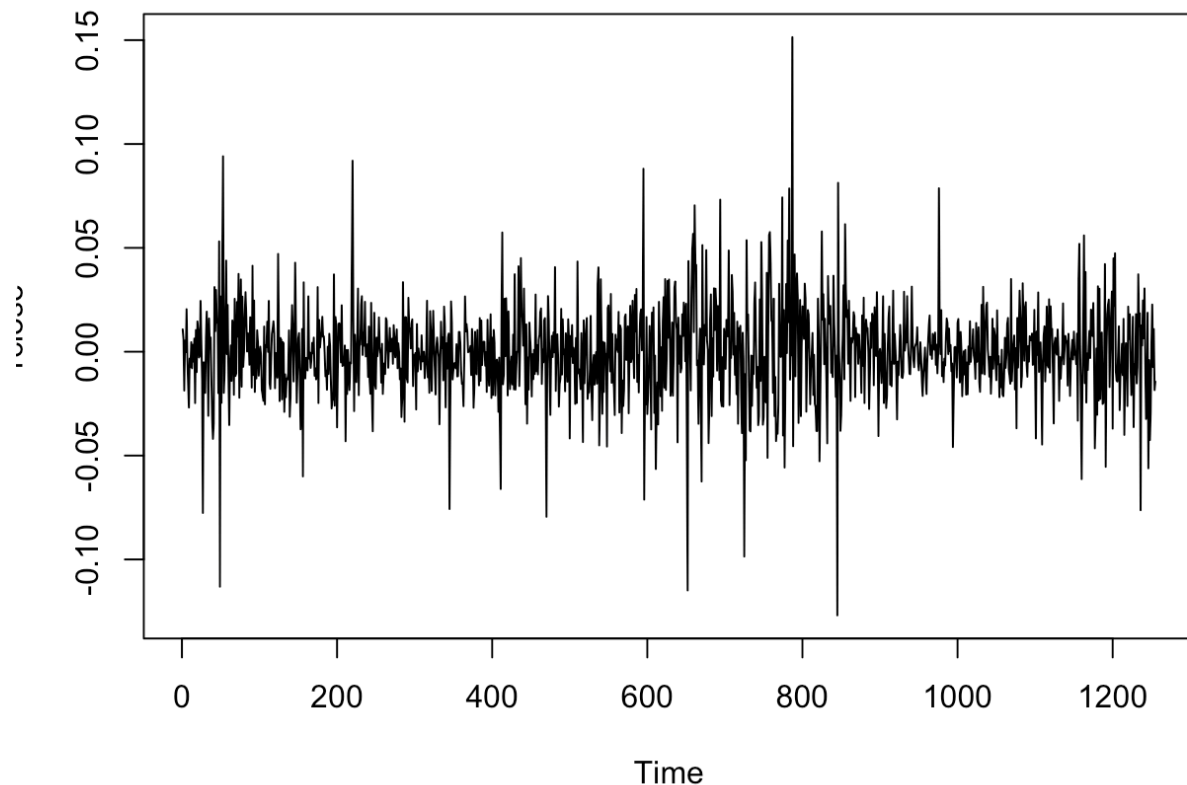
Since:

The p-value is 0.6439, which is much higher than 0.05,

We fail to reject the null hypothesis of the ADF test (which is: the series has a unit root),

Therefore, the series is non-stationary.

Log Return:



Code:

```
rclose=diff(log(Close))
> plot.ts(rclose)
```

Link: <https://www.youtube.com/watch?v=ny3gRhVsi4>

Second Stationary Test (Dickey Fuller Test):

Augmented Dickey-Fuller Test: This time the test is stationary as we have a p-value which is below or greater 0.05.

data: rclose

Dickey-Fuller = -11.087, Lag order = 10, p-value = 0.01

alternative hypothesis: stationary

In `adf.test(rclose)` : p-value smaller than printed p-value

Series: rclose

ARIMA(0,0,0) with zero mean

sigma^2 = 0.0005053: log likelihood = 2982.13  
AIC=-5962.27 AICc=-5962.26 BIC=-5957.13

Results from model rclose:

Call:

arima(x = rclose, order = c(0, 0, 0))

Coefficients:

intercept

-4e-04

s.e. 6e-04

sigma^2 estimated as 0.0005052: log likelihood = 2982.3, aic = -5960.6